

In This Special Issue: When Attention Is All Marketers Need—Artificial Intelligence in Marketing

I. INTRODUCTION

ARTIFICIAL Intelligence (AI) is having a transformative impact on business and society [1]. Marketing is likely the business function that has been impacted first and most by the recent developments in machine learning and AI technology. Ubiquitous examples include algorithmic targeting for advertising, chatbots for customer service, and natural language processing/ customer sentiment analysis for marketing research [2], [3]. Rapid improvements in transformer models, especially the algorithms powering generative AI applications like ChatGPT and DeepSeek, have surprised even experts. They demonstrated that attention-focused machine learning algorithms can perform cognitive tasks that previously were thought to require more complex processes. The title of our Special Issue Editorial, paraphrases that of the seminal article on transformer models—“Attention Is All You Need”—to suggest that marketers may be able to deploy AI in an ever-increasing number of tasks [1]. This development is both exciting and bewildering to every marketing professional with whom we have had contact.

II. AI IN CONSUMER-FOCUSED LITERATURE

We recently published an exhaustive analysis on, among other things, how we got to where we find ourselves in the AI world today [4]. We investigated references to AI and related methods used in over 2,200 scientific papers published between 1955 and 2020 in the leading fifty academic journals across all areas of marketing-interfacing management research, which examined consumption-related questions. Using topic modeling, we examined how the use of AI in consumer-focused academic literature has evolved over the years, to ask: Which consumer-relevant topics have emerged in the use of AI techniques and have been prioritized? Which topics and techniques offer opportunities for future development? Our work highlights productive consumer-relevant applications of AI techniques. We do this in two ways: (1) at the conjunction of *technique* and *topic*. For several prominent AI techniques, we identify concrete opportunities based on the method’s strengths; and (2) at the conjunction of *topic* and *theory*. We match the consumer topics emerging from our topic modeling analysis with the consumer psychology of brands, to understand the scope and focus of current developments.

Our topic modeling analysis arrived at 16 consumer topics ranging from brand communities to consumer preferences,

to effects on firms, and; 16 AI technique topics that are grouped into specific types of algorithms (e.g., support vector machines) or broader application domains (e.g., image processing). We found that over 90% of consumer-relevant research using AI techniques has taken place since 2009. The results point to the present day developments and, especially, the future of marketing application of AI techniques [5]. In this Editorial we offer a few examples that demonstrate the transformative power of AI in helping companies understand and serve customers. At the same time, AI raises a number of challenges, which we also discuss. These challenges relate to its practicality, claims, and ethics.

III. MARKETING RESEARCH: SIMULATING CUSTOMERS FOR MARKETING INSIGHTS

A top contender for most exciting new AI-marketing interface is the use of Generative AI for consumer insight generation. AI firms like Cohere, instead of relying on real consumers and their human-made data, are now generating data from computers to train AI models. This computer-made data is a novel and cost-effective pathway to acquire terabytes of data based on “synthetic consumers”. By creating synthetic data customized to specific consumer contexts (e.g., online purchase interactions, product evaluation), Cohere is drawing on scalable language systems to generate synthetic text that mimics semantic similarities, classification or paraphrasing of text by real consumers [6]. Relative to real world data, such synthetic data can, without getting entangled in privacy and bias problems, simulate a variety of consumer contexts, and do so speedily, in large amounts and at low cost.

In parallel, academic marketing is leveraging Large Language Models (LLMs) as a “synthetic consumer” by deploying their quick text generation features to prompt survey responses and investigate customers’ preferences [7]. This way LLMs are being used to generate a distribution of consumer responses rather than a single response. Using OpenAI’s public API, marketing researchers can use GPT to simulate human responses by submitting a range of prompts and rapidly generate a distribution of responses per prompt, thereby reflecting preferences of consumers. Even though the results can be sensitive to order, phrasing, and other prompt details, responses generally align with both actual consumer data and economic theory. Marketing scholarship is also making significant progress in creating synthetic expertise. Such a scalable, independent “synthetic-expert” uses generative AI to serve as a proxy for an expert human agent who can parse through unstructured text (social media, news reports

etc.) to correctly label complex marketing constructs (e.g., 4Ps: product, promotion, price, place) that hitherto were labeled mostly by human domain experts. Such a “synthetic-expert” should find diverse applications in, for example, legal (case classification), health (patient diagnoses), and industrial (technical specifications) contexts [8].

IV. MARKETING STRATEGY: AI FOR SEGMENTATION, TARGETING, AND POSITIONING

One of the oldest problems in marketing is precision in segmentation, targeting and positioning, which are the crux of marketing strategy [9]. One of our key methods-based topic models on AI techniques explores how AI can help firms solve classic problems as they learn about their segments, while grouping demographic, product usage and self-reported preferences. Another topic model investigates how customers can be grouped to determine items purchased, purchase preferences and recommendations. Broadly associated with problem solving and learning methodological toolboxes, these topic models have been used by academics to predict consumer choice and diversity of users. Some firms attempting to capitalize on AI techniques in this way include P&G, JP Morgan Chase, and Allen & Overy.

A. Case 1: Proctor & Gamble

Consumer-goods behemoth Procter & Gamble (P&G) is experimenting with an AI patent that helps classify consumers’ photos based on sub-pixel-level dimensions of their hair and skin tones. P&G’s focus on such AI-powered beauty tech brings sharper focus on meaningful groups of buyers of products in hair care and UV protection, while keeping the focus on inclusivity [10]. However, marketing managers at P&G also realize that sharply defined segments must be backed up by scalable manufacturing that is not only smart but timely. By embracing an AI-based digital manufacturing platform P&G aims to align instantaneous product quality inspections with specific needs of a consumer segment [11].

B. Case 2: JPMorgan Chase

JPMorgan Chase is going all in on identifying profitable segments by seeking a patent for its AI-based investment advisor – IndexGPT [12]. The AI-based advice augments human resources through “temporary use of on-line non-downloadable cloud computing software using artificial intelligence for use in computer software selection of financial securities and financial assets” [13]. It is a big leap for financial services firms to use AI for “analyzing and selecting securities tailored to customer needs” [14]. Having topped the first AI index for banks [15], JP Morgan is likely to deploy its cutting-edge patented investment advisor in a suite of tools ranging from advertising and marketing services, to target insurance and financial products tailored to different investor segments spanning from investment portfolio clients to retirees [15].

C. Case 3: Allen & Overy

It is not just industry giants. Even legal service providers are now veering towards AI-based legal innovation to secure

a competitive advantage for their offerings. While complex legal technicalities are still some time away, AI-based tools are being leveraged for a distinctive strategic positioning. For example, the legal offices of Allen & Overy now rely on Harvey AI; a self-service platform that aligns processes aimed at improving client subscriptions of investment funds [16]. Harvey AI’s intersection with technology and law is unique, and so is its positioning as “generative A.I. for elite law firms” that can “work in multiple languages and across diverse practice areas, delivering unprecedented efficiency and intelligence”. Allen & Overy expect better targeting of clients by achieving more efficient regulatory compliance, backed by a toolbox of due diligence, contract analysis, and litigation [17].

D. On the Limits of Futurecasting When Using LLMs

One caveat is that AI-driven models for marketing and consumer choice rely on past data to predict future behavior, a very useful way of examining large datasets and fine-tuning market segmentation. However, this retrospective approach inherently limits their ability to anticipate paradigm-shifting innovations. As Steve Jobs noted [18], true innovation is about understanding what consumers will want before they realize it themselves. Similarly, Henry Ford’s famous remark “*If I had asked people what they wanted, they would have said faster horses*”, underscores how customer preferences, if solely based on past experiences, may not capture groundbreaking advancements. AI excels at pattern recognition and optimizing existing trends, but it struggles with visionary leaps that redefine markets. If AI had been tasked with forecasting Kodak’s future 20 years ago, it likely would have refined film-based strategies rather than recognizing the impending dominance of digital photography, a paradigm shift that caused that company to fail.

The fundamental limitation lies in AI’s reliance on historical data and probabilistic modeling, which can reinforce existing market assumptions rather than challenge them. AI can fine-tune consumer targeting and enhance efficiency within known frameworks, but it lacks the creative intuition to generate disruptive ideas that reshape industries. True innovation often comes from human ingenuity [19], which defies patterns, takes risks, and envisions possibilities beyond the scope of available data. While AI can enhance decision-making, it remains ill-equipped to predict or drive the kind of radical shifts that have historically defined market revolutions (ironically, perhaps, even the impact of AI itself).

V. CHALLENGES OF AI: PRACTICALITY, CLAIMS AND ETHICS

A. The Practicality of AI Solutions

The current hype around AI, and generative AI in particular, risks getting business leaders to underestimate the design and implementation issues. Despite the recent leaps forward, several practical challenges remain. For example, going back to the use of generative AI to simulate marketing research respondents, some researchers are questioning the validity of LLMs in consumer settings [20]. While some GPT models have been found to be generally in alignment with foundational

economics (e.g., risk aversion and demand curves), academics are still validating the accuracy of the insights that can be generated by synthetic respondents. Moreover, concerns have been raised around prohibitive costs and strained data infrastructures due to the diversity of data types and computational complexities [21]. If one adds the difficulty of optimizing consumer interfaces, matters are only likely to get even more complex. Surveys reveal that firm-wide applications that would drastically minimize barriers and benefit many customers are still way off. Today, one third of AI projects are still in pilot stages, while eight in ten face technical barriers such as those related to data management, security issues, computing power [22]. At the same time, many companies are struggling to articulate effective management processes, and even to define emerging organizational roles in the areas of data and analytics [23].

B. AI Claims, Ambiguity and Confusion

To make matters worse, AI claims are catching the attention of the Federal Trade Commission (FTC) that has had to nudge marketers to “Keep your AI claims in check” [24]. Calling AI, a “marketing term”, and “a hot one” too, the FTC’s guidance alerts consumers to its overuse or even abuse by some advertisers. The FTC has warned consumers that “some products with AI claims might not even work as advertised in the first place” [24]. In a 2022 report to the U.S. Congress, the FTC was of the view that AI was an ambiguous term and that it had been making efforts to inform marketers that the use of AI in advertising may trigger the FTC’s framework. Among other things, the FTC framework focuses on: the exaggeration of claims, the promise of a better product, risks, and whether the product even uses AI [25].

Our systematic analysis of academic publications highlights the ambiguous definition of AI [4]. The development of AI has progressed nonlinearly over the past few decades. Take for instance, AI models that seemed sophisticated at one time, fell out of favor at a later date only to be revived more recently (e.g., neural networks). Unsurprisingly, the most commonly used statistical techniques in publications about AI turns out not to be deep learning or some other machine learning method, but variations of the humble regression analysis, a methodological workhorse which has been around for a very long time. In the absence of accepted definitions and conventions, the scope of the AI phenomenon remains blurry. This makes consumers vulnerable to confusion and exploitation.

C. AI Ethics and the Attention Economy

Beyond the issue of consumer disinformation, ethical issues abound. In some cases, AI techniques do appear to sanction what some fear are potentially exploitative practices, or at least practices that might be deleterious to consumer welfare. One of these relates to an age-old marketing problem: finding the best way to attract consumers’ attention. Our research shows that the approach taken by some firms to build consumer-brand relationships can set off alarm bells. AI models such as K-Nearest Neighbor can be used to develop new insights through real time predictions by tracking ads [26]. But such

tracking can be invasive to consumer privacy and even be misused to profile unsuspecting consumers in ways they would not approve. In general, the “attention economy” prizes engagement above all else. The trove of data about consumers that firms can collect by tracking online footprints is used to power the algorithms that regulate exposure to all sorts of content online. These algorithms are often optimized for goals that may not be aligned with consumers’ best interest, often undermining well-being and productivity [44].

D. AI, Trust and Behavioral Engineering

In other cases, consumers can perceive unfair treatment or feel that their trust is being misused when they realize that firms are putting in place processes to understand how a consumer’s brain generates patterns of behavior. There are several concerns at play here. First, like in the case of consumer attention discussed above, there are consumer privacy issues. Second, consumer welfare issues arise if predictive analytics are used for potentially exploitative goals [27]. While machine learning has been used to make predictions about consumer behavior for over two decades now, newer techniques can be much more effective [28]. For example, our review of the academic literature [4] shows how deep learning models can be used effectively to trigger impulse purchases, which may not be in line with—and in many cases in fact be directly in opposition to—consumers’ long-term goals.

Learning about such behavioral patterns has been among the oldest problems not only in marketing but in business more generally. AI-powered prediction models can have darker sides as they can help firms mine data about, for example, eye movement and negotiation behavior [29]. Because these models are trained on data embedded with existing social and cultural biases found on the internet that are unrepresentative datasets, their outcomes may also reflect and reinforce discriminatory patterns—potentially disadvantaging particular subgroups, such as racial, religious, and ethnic minorities. Algorithmic bias in ML-based marketing models can result in unfair outcomes for customers, such as racial bias in medical algorithms that serve patients, bias in ad targeting for executive leadership roles, and even expensive taxi rides for those who frequent destinations with an ethnic population [26, pp. 201–202]. Algorithmic bias has socio-ethical implications. Deployment of black-box algorithms is also likely to make many buyers (and policy makers) worried about misinformation and trickery [30].

Recent studies reveal anxieties about the spread of AI impacting overall well-being through deepfakes, surveillance digital technologies, and harassment [31]. Similarly, contemporary developments in consumer location tracking allow for consumer push services in physical shopping malls [32], while computer vision allow quantifying patterns of behavior through a technique called pose estimation [29]. While this technique may lower prices as it saves time, it has also been found to be intrusive, almost as meddlesome as computer-based motor tracking studies. Since pose estimation (e.g., SLEAP [33], DeepLabCut [34]) spots body joints in a video and outputs them as coordinates, the versatility of the framework may encourage misuse. Some firms may

capitalize on infrared-based tracking of consumers and the data generated. Such pose estimation can be looped back for surveillance of consumer tastes and client experiences, described as forming the underpinnings of retail uberveillance theory [35], [36], [37].

This review of AI challenges is not meant to serve as a call for suspending the development of AI tools, or for prohibiting their adoption by firms. Business needs innovation to better serve customers, promote economic growth, and ultimately improve quality of life. As in the case of other new technologies that were initially unevenly disseminated and poorly understood, the point instead is that with power comes responsibility. Firms that use AI responsibly have an interest in ensuring that the technology is deployed to the benefit of long-term business goals. These are poorly served by triggering consumer backlash or regulatory straitjackets, and firms have therefore an interest in fostering trust in AI and consumer acceptance.

VI. CONCLUSION

In the examples above we focused on how AI can help solve old marketing problems in innovative ways. At the same time, new marketing problems are also emerging. It is hard to provide a confident inventory because the field is changing so fast. But one way to look for emerging problems is to ask whether these questions propel marketing interfaces forward. That is, do these discussions edge on the frontiers of marketing as consumers begin to interact with “new technologies and rapidly emerging business models” [38, p. 1]. As traditional boundaries between marketing and adjacent disciplines get blurred, marketing researchers are increasingly posing questions and making contributions across the once rigid aisles ranging from computer science, statistics and economics to information science and operations. Such scholarship is only reflective of the fluid times we live in.

The emergence of new interactive wearables [39] (such as, smartphones, smartwatches, two-way digital glasses, smart headbands that track brainwaves, metaverse immersive goggles, Internet of Things devices, and even implantables) will continue to alter the marketing interface. Together with AI-enabled computer audio-visual monitoring to detect consumer knowledge levels, mood and emotion using biometrics, emerging technologies will cause a redefinition of marketing as we know it. Whereas in the past, we might have asked the question does more data about a consumer mean greater accuracy about their predicted behavior, e.g., buying patterns and the like. Today, the data emanating almost directly from the user, via keyboard input, or voice or social media, has meant that companies know even more about their customers, before, during and after the buying process. While much has been claimed about the buyer’s agency still being in-tact, the onslaught of micro-messaging customized to consumer patterns ‘at just the right time’, has made stickiness drivers plausible, and lock-in a possibility [40].

WHERE ARE WE HEADED?

ElevenLabs [41] has been dabbling in synthetic voice since 2022. As a voice technology company, ElevenLabs is using AI to mimic emotional prosody and pitch variation

generating voice content that can target a plethora of marketing interfacing settings, such as customer service, and accessible products targeting consumers with voice health complications. Such voice cloning can also be used to make “content universally accessible in any language and voice”, thereby, strengthening connections between consumers and brands that may want to rely on synthetic voices of brand ambassadors, among others [41]. Another interesting disrupter is Asset Entities <https://aeai.me/>. This is an instance of AI systems showing up rapidly in Web3’s decentralized marketing interfaces. Here, companies such as Asset Entities are capitalizing on online ecosystems to sell digitally identified NFTs based on AI. Highly context-dependent, such product lines could, for example, create products that “celebrate the amazing weekend Box Office performance of Barbie” [42], and sell special editions of AI bots that are then uniquely owned by users.

Our research corroborates recent industry insights that found gross under usage of cutting-edge AI techniques. Going back to our editorial title, one can now see a different meaning in it. As our research shows, the techniques are there, they only need to be deployed to solve marketing problems. Paying attention and exploring what is already available today is all marketers need.

IN THIS SPECIAL ISSUE

This special issue consists of 8 papers on the main theme of “AI, Empathy and People”. The first paper in the special acts as a framing within which to consider the role of empathy in the context of AI for the collection of papers presented in this issue. The paper addresses IEEE 7014-2024 [43], an IEEE Standard for Ethical Considerations in Emulated Empathy in Autonomous and Intelligent Systems. Professor Andrew McStay of Bangor University in the U.K. has written specifically on IEEE P7014.1™ *Recommended Practice for Ethical Considerations of Emulated Empathy in Partner-based General-Purpose Artificial Intelligence Systems*. Article [A1] provides a way forward for all new applications in services where AI interacts with people. This standard defines “a model for ethical considerations and practices in the design, creation, and use of empathic technology, incorporating systems that have the capacity to identify, quantify, respond to, or simulate affective states, such as emotions and cognitive states. This includes coverage of “affective computing,” “emotion artificial intelligence,” and related fields” [43].

The second paper [A2] is written by Jordan Tschida, Katina Michael and Troy McDaniel from the School of Computing and Augmented Intelligence, as well as the School for the Future of Innovation in Society and the School of Manufacturing Systems and Networks at Arizona State University. The work has been supported in part by two NSF grants and in part a Zimin Institute competitive grant. The paper explores “Social Robots for Healthy Older Adults”, in the hope of overcoming loneliness and social isolation, conditions that are associated with adverse health outcomes. The study employed a Wizard-of-Oz experimental design to investigate verbal and nonverbal communication levels over a 4-week period. Data was collected from the 16 participants in

three phases beginning with the human-to-robot conversation observation immediately followed by a post-interaction survey, open-ended interviews, and finally a post-experience survey. Participants reported positive experiences with the robot, including companionship, enjoyment, and emotional support.

The third paper [A3] is written by Raffaele Ciriello, Angelina Chen, and Zara Rubinsztein from the University of Sydney Business School. It provides some guidance for future companion-related software (e.g., chatbots) and hardware (e.g., robots). The paper's contribution is in presenting a framework for compassionate AI design, emphasizing equitable distribution of benefits and burdens based on stakeholder vulnerability. The paper prioritizes empathy and dignity to guide responsible AI development. The authors distinguish between human empathy and emulated empathy. They also point to recent cases such as the suicide of Sewell Setzer III after interactions with a Character.AI chatbot and Jaswant Singh Chail's assassination attempt following Replika chatbot support, highlighting the dangers of unregulated AI. The work uses Arthur Schopenhauer's compassionate imperative as a plausible ethical framework to apply in context.

The fourth paper [A4] is co-authored by philosopher Evan Selinger of Rochester Institute of Technology and MD Thomas Carroll at the University of Rochester. The paper uses three scenarios to unpack levels of empathetic medicine in practice. These include: (1) AI-powered "doctor" avatars directly interacting with patients, (2) AI editors and co-authors helping clinicians find the right words, and (3) AI simulated patients helping to provide communication training. The paper emphasizes that the expression of empathy is an important part of effective and humane medical care, and that doctors today face daily challenges, at least in part due to the ever-increasing demands on clinicians' time. In each case, the authors identify fundamental AI and medical ethics considerations, including those that warrant further research. This paper has wide-reaching implications for care management using AI.

The fifth paper [A5] is written by Szu-Yu Kuo of the National Kaohsiung University of Science and Technology and Liang-Bi Chen of the National Penghu University of Science and Technology in Taiwan. This paper examines the relationship among emotional intelligence, artificial intelligence, and safety behavior in the context of container terminal operations. The findings indicate that both emotional intelligence and AI have a significant impact on safety behavior on a port. The research was supported in part by the National Science and Technology Council (NSTC), Taiwan, under Grant NSTC 113-2410-H-992-049 and Grant NSTC 112-2221-E-346-002-MY2.

The sixth paper [A6] focuses on the Future of Work and is written by Elma Hajric, Farah Najar Arevalo, Leonard Bruce, Fritz Antony Smith, and Katina Michael from Arizona State University, funded in part by a National Science Foundation - National Research Training grant #1828010 on Citizen-Centered Smart Cities and Smart Living. This article considers a human resource management (HRM) workplace scenario where employees are monitored through cameras on personal electronic devices for the purposes of facial emotion recognition. The applications described pertain broadly to the future of work context. The article considers how employers, might use

employee facial emotion data for data-driven decision-making. It concludes by identifying the social implications and policy recommendations of facial emotion recognition in a future of work context, providing a list of do's and don'ts in human-centered values, inclusive of confidentiality, accuracy, fairness, safety, and transparency.

The seventh paper is co-authored by Christos Bormpotsis of EmotiZen, with Michael Nanos of the University Hospital of Regensburg and Asma Patel from the Aston Business School. Together they have written [A7]. The paper delves into financial decision-making which is a complex cognitive process concerned with emotional state and behavioral bias. The paper aims to decode the neural mechanisms behind financial behavior to advance theoretical and empirical progress in neurofinance. The societal implications of this research, seeks to encourage equitable, stable and informed financial systems while addressing challenges at the intersection of neurofinance and neuroscience-informed AI.

The eighth and final paper in this special issue is written by Marta Beltrán who is currently the Chief of the Scientific Unit with AEPD, the Spanish Data Protection Authority. Article [A8] gets to the heart of the potential for AI in everyday digital consumer products and services. The proposed FoSIP framework (Forced action, Social engineering, Interface interference, Persistence) offers a means by which innovators can improve product design providing a vehicle for autonomous decision-making by consumers. Beltrán makes the case that our digital products and services must espouse empathy as a value by default, and that future services are set to become even more complex. Thus, if we are able to predict how particular services may make a consumer or user behave then we have a commensurate responsibility not to abuse this behavioral knowledge toward their financial exploitation. Associate Professor Beltrán is presently on leave from the Universidad Rey Juan Carlos, where she researches distributed systems, cybersecurity and privacy.

APPENDIX: RELATED ARTICLES

- [A1] A. McStay, "Emulated empathy: Can risks be countered by a soft-law standard?" *IEEE Trans. Technol. Soc.*, vol. 6, no. 3, pp. 250–256, Sep. 2025, doi: [10.1109/TCYB.2021.3062949](https://doi.org/10.1109/TCYB.2021.3062949).
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- [A5] S.-Y. Kuo and L.-B. Chen, "Utilizing emotional intelligence and artificial intelligence to improve safety behavior in smart port operations," *IEEE Trans. Technol. Soc.*, vol. 6, no. 3, pp. 283–294, Sep. 2025, doi: [10.1109/TTS.2024.3513775](https://doi.org/10.1109/TTS.2024.3513775).
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- [A8] M. Beltrán, "Defining, classifying and identifying addictive patterns in digital products," *IEEE Trans. Technol. Soc.*, vol. 6, no. 3, pp. 314–323, Sep. 2025, doi: [10.1109/TTS.2025.3564840](https://doi.org/10.1109/TTS.2025.3564840).

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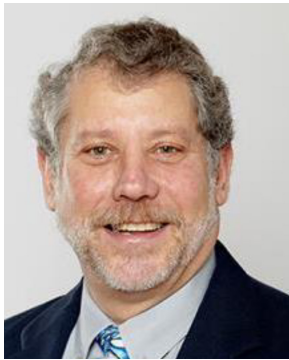
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